

AI FAQ ASSISTANT API

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PROJECT DESIGN PHASE

(AI FAQ ASSISTANT API)

The **Project Design Phase** defines the overall structure, components, data flow, and functionality of the AI FAQ Assistant API. The system is designed to provide users with quick and relevant answers to frequently asked questions through an AI-based API.

1. System Architecture

The system follows a **client-server architecture**. The user sends a question through the application, and the request is forwarded to the backend API. The API processes the request, interacts with the FAQ database and AI module, and returns a suitable answer to the user.

2. Main Components

- **User Interface:** Allows users to enter questions and view answers.

- **API Server:** Handles requests and manages communication between the application, database, and AI module.
- **AI FAQ Module:** Understands user questions and provides relevant answers.
- **Database:** Stores FAQ questions, answers, user details, and related information.
- **Authentication Module:** Handles user registration and login securely.
- **Admin Module:** Allows authorized users to add, update, and delete FAQ content.

3. API Design

The API follows REST principles and provides separate endpoints for different operations.

Example endpoints:

- `/api/user/register` – Register a new user
- `/api/user/login` – Login user
- `/api/faq/create` – Add a new FAQ
- `/api/faq` – View FAQ information
- `/api/faq/:id` – View or update a specific FAQ
- `/api/ask` – Send a question and receive an AI-generated answer

4. Data Flow

The basic data flow is: **User enters a question** → **API receives the request** → **Request is validated and processed** → **AI FAQ module identifies relevant information** → **Answer is generated** → **Response is returned to the user.**

5. Security Design

The system includes authentication and authorization to protect user and administrative functions. Input validation is used to handle invalid requests, while sensitive user information is stored securely.

6. Design Goal

The main goal of the design is to create an **easy-to-use, secure, scalable, and efficient AI FAQ Assistant API** that can provide relevant answers and allow FAQ information to be managed easily.

➤ Conclusion

The Project Design Phase provides a clear blueprint for developing the AI FAQ Assistant API. By combining the API server, database, authentication, and AI FAQ module, the system

can efficiently handle user questions and deliver relevant responses. This design also provides a strong foundation for the development and implementation phases of the project.